

Chapter 10: Metal extraction



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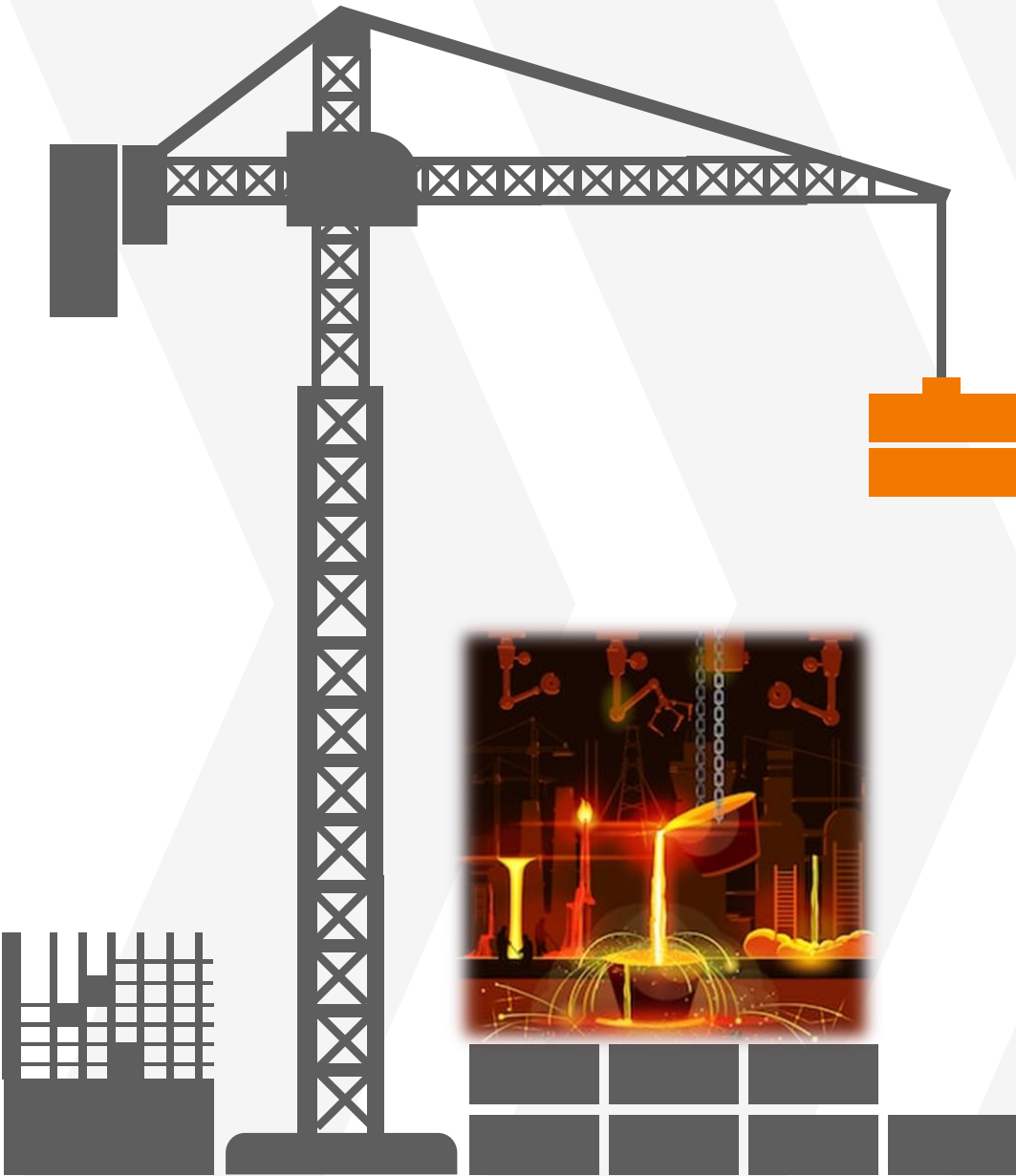
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Introduction

❑ Extractive metallurgy

Extractive metallurgy is the practice of removing valuable metals from an ore and refining the extracted raw metals into a purer form. In order to convert a **metal oxide** or **sulfide** to a **purer metal**, the ore must be **reduced physically, chemically, or electrolytically**.



Introduction

- ❑ Metals are extracted from ores. An ore is a rock that contains enough of a mineral (metal compound) for the metal to be extracted from it. Most metals are extracted from an ore by reduction with carbon or by electrolysis.
- ❑ **Metal extraction process** can be divided into three main stages.
 - Ore preparation
 - Extraction
 - Purification of metals

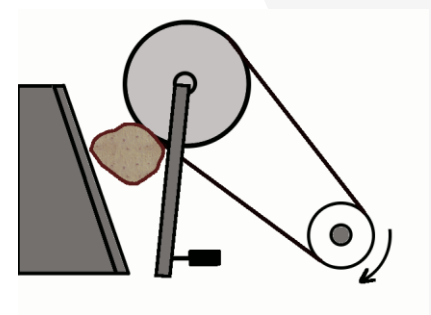


Ore Preparation

□ In the field of extractive metallurgy, mineral processing, also known as **ore dressing**, is the process of separating commercially valuable minerals from their ores.

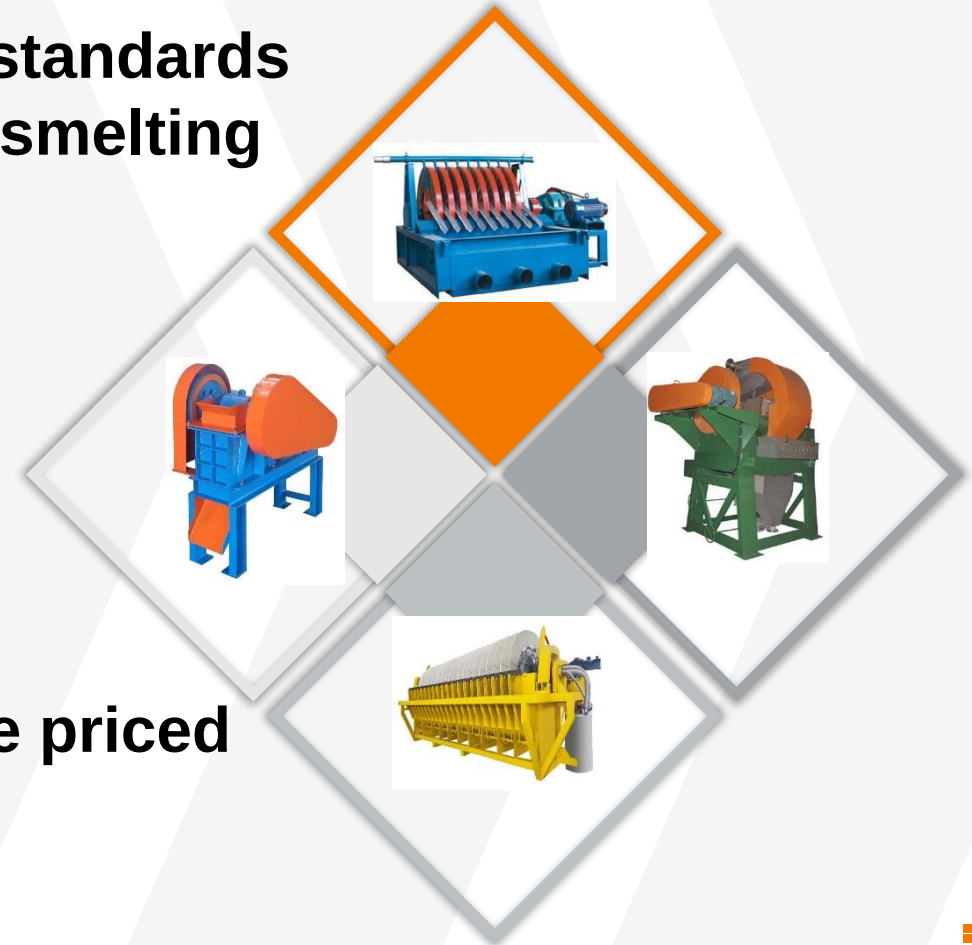
□ It takes multiple steps to extract the **"important"** element from the ore:

- **First**, the minerals need to be crushed and grinded from unwanted rocks.
- **Then**, the ore must be screened and classified.
- **Finally**, the minerals need to be concentrated and separated of the ore.



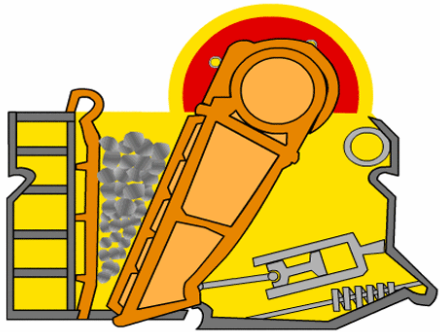
The purpose of ore dressing

- ❑ Make a high concentration of ores, standards required by the market, suitable for smelting and energy saving for smelting.
- ❑ Save cost of the transportation.
- ❑ Reduce metal in slag in smelting process.
- ❑ Can separate other minerals that are priced to sell, or continue to use it.



Ore dressing machine

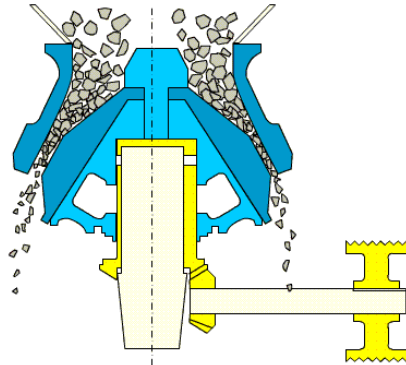
Primary crushing



Jaw crusher

Reduced large rocks into smaller rocks, gravel

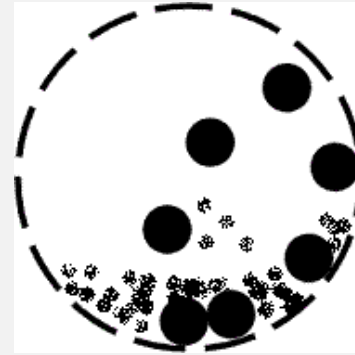
Secondary crushing



Cone crusher

Grinding in which the discharge from the primary crusher is broken down to a size suitable for feed to fine grinding machine

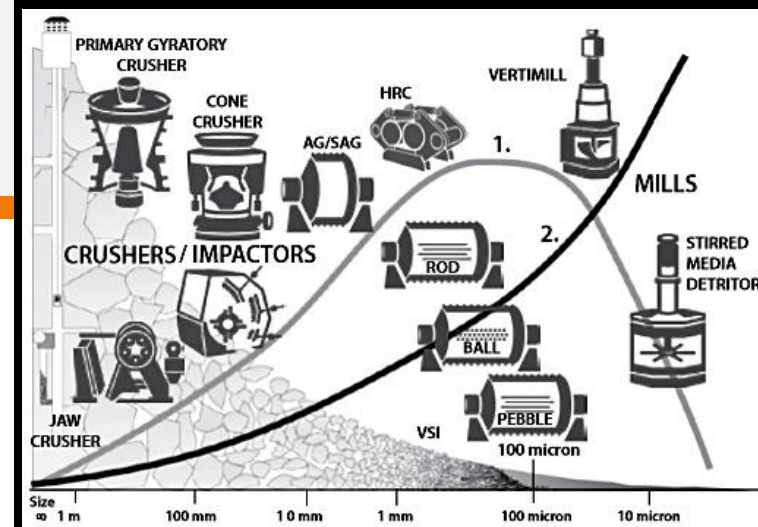
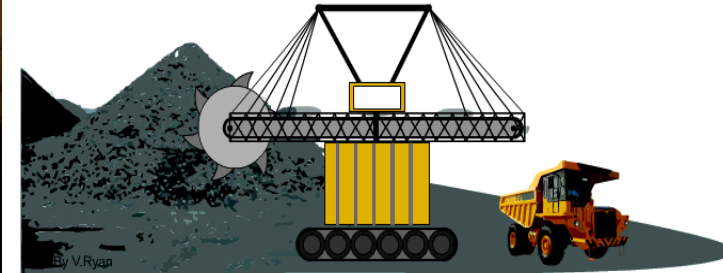
Tertiary crushing or Grinding



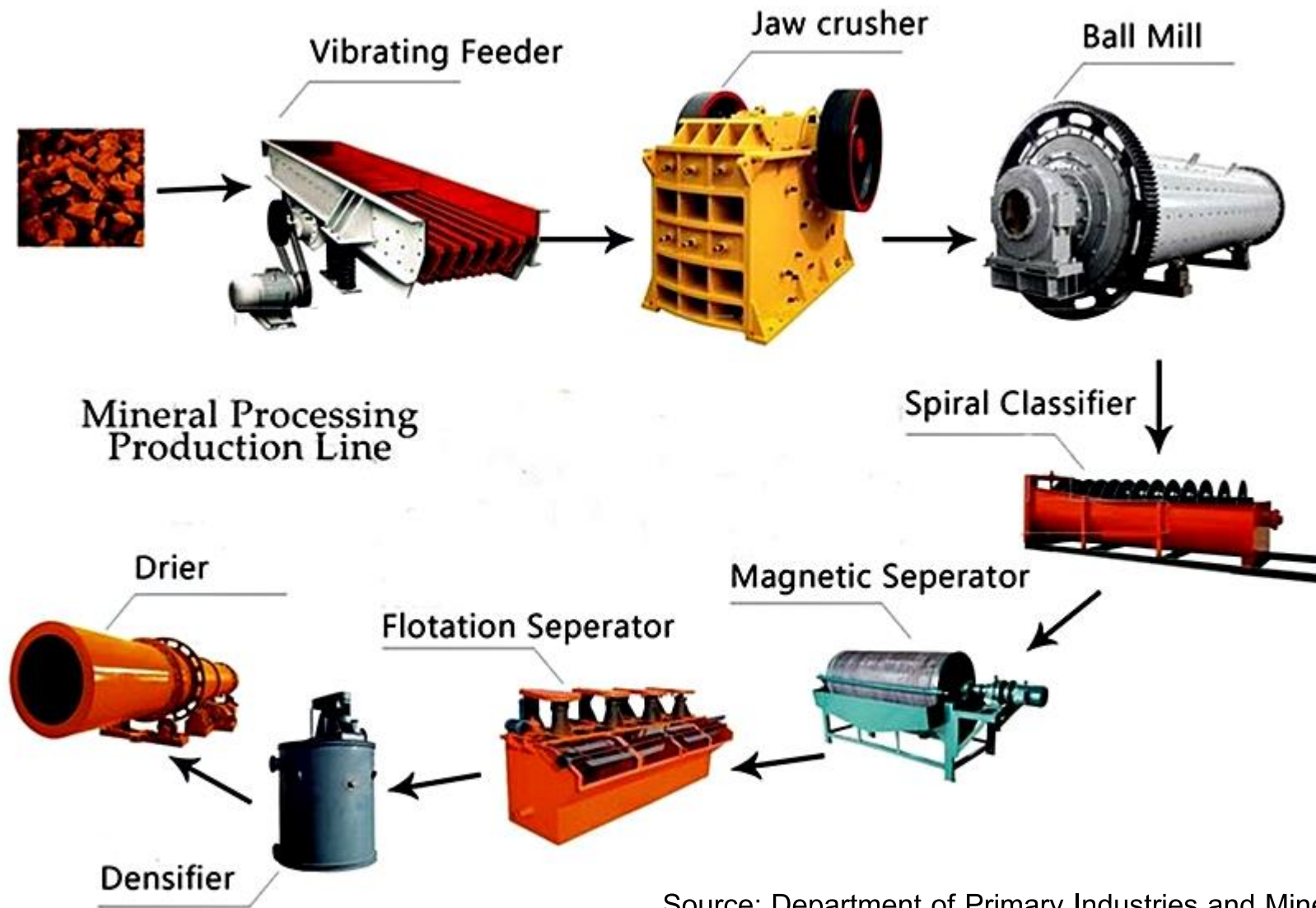
Ball mill

Size reduction by crushing has a size limitation for the final products. If we require further reduction, say below 5-20 mm

2. STORED AT MINE SURFACE - REMOVED FOR PROCESSING



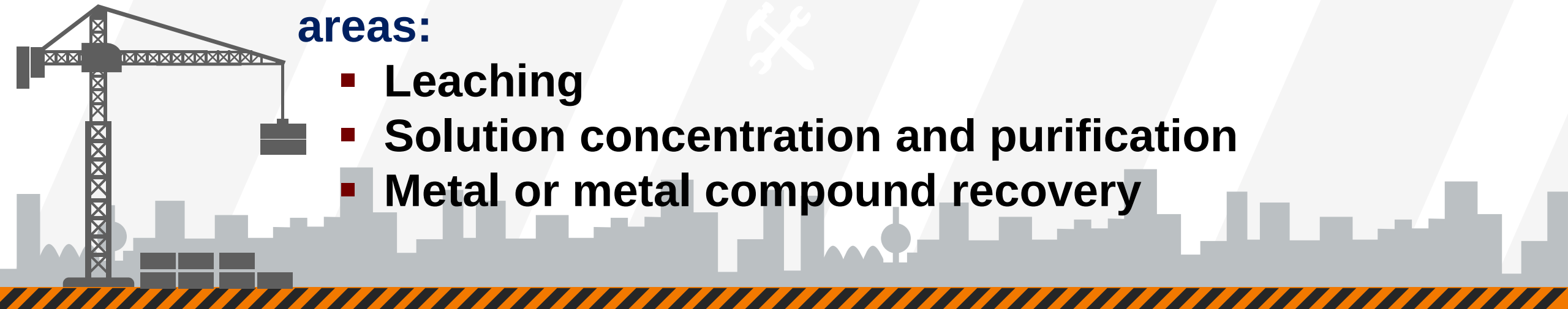
Mineral Processing



Source: Department of Primary Industries and Mines.

Metal Extraction

- ❑ **Hydrometallurgy** is a method for obtaining metals from their ores. It is a technique within the field of extractive metallurgy involving the use of **aqueous chemistry** for the recovery of metals from ores, concentrates, and recycled or residual materials.
- ❑ **Hydrometallurgy** is typically divided into three general areas:
 - Leaching
 - Solution concentration and purification
 - Metal or metal compound recovery





Leaching

Leaching is a process where ore is soluble and impurities are insoluble, widely used **extractive metallurgy** technique which converts metals into soluble salts in aqueous media.

Compared to pyrometallurgical operations

- Leaching is easier to perform and much less harmful, because no gaseous pollution occurs.
- Drawbacks of leaching are the highly acidic and in some cases toxic residual effluent.

Leaching is done in long pressure vessels which are cylindrical (horizontal or vertical) or of horizontal tube form known as **autoclaves**. A good example of the autoclave leach process can also be found in the metallurgy of **zinc**. It is best described by the following chemical reaction:



This reaction proceeds at temperatures above the **boiling point** of water, thus creating **a vapor pressure** inside the vessel. Oxygen is injected under pressure, making the total pressure in the autoclave more than **0.6 MPa** and temperature at **473-523 K**.

Leaching agents

The choice of a leaching agent depends on many factors:

01

Chemical and physical character of the material to be leached.

03

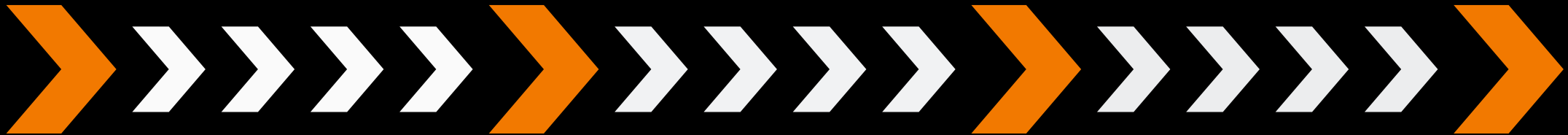
Selectivity for the desired constituent to be leached.

02

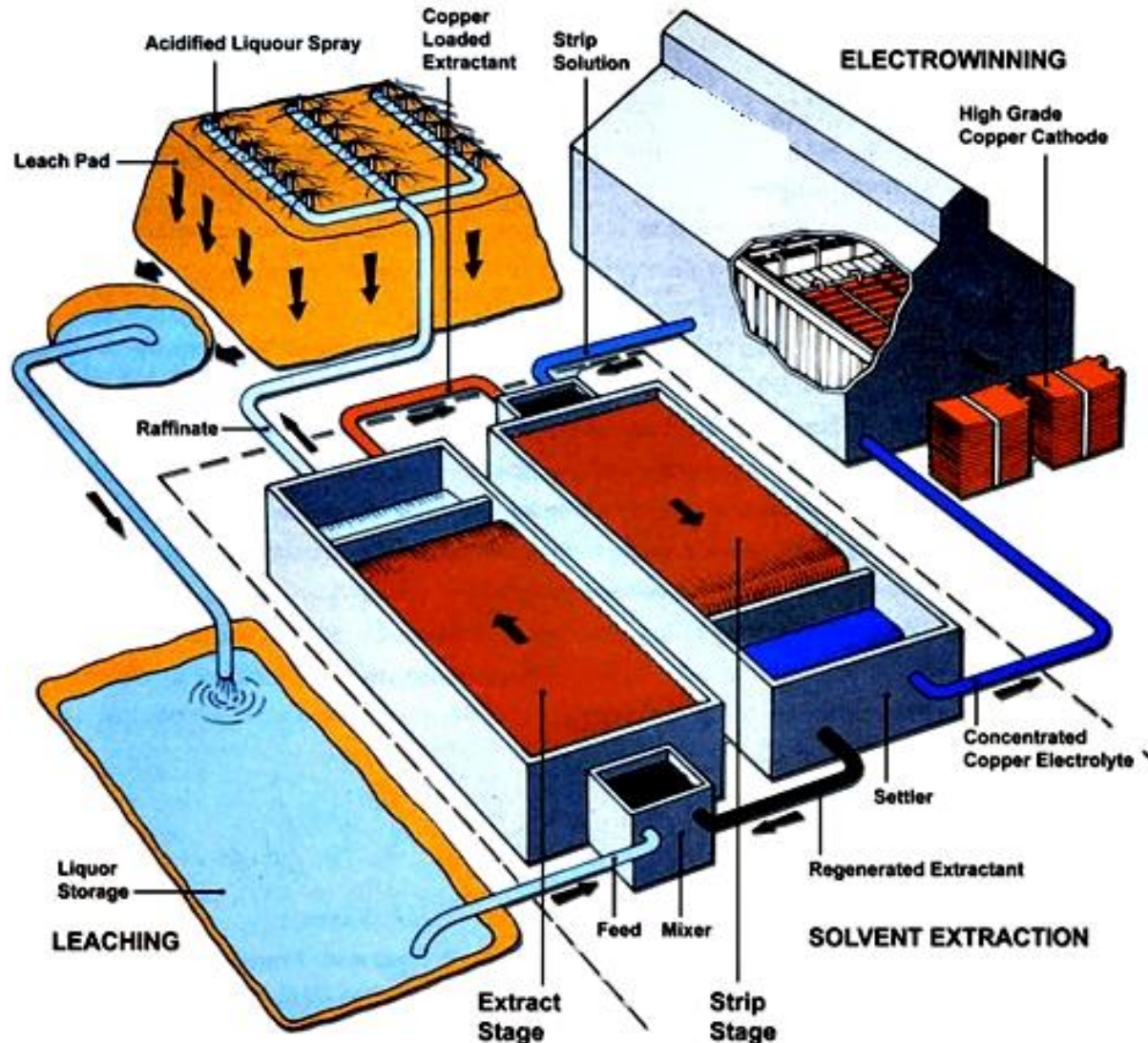
Cost of the reagent.

04

Ability to be regenerated, e.g., in the leaching of ZnO by H_2SO_4 the acid is regenerated during electrolysis.



Metal Extraction



Hydrometallurgy

Leaching

- Puts copper in an acid water solution

Solvent Extraction

- Transfer copper from leach to electrowinning
- Transfer acid from electrowinning to leach

Electrowinning

- Transfers copper from solution to metal

Leaching



Solvent Extraction



Electrowinning

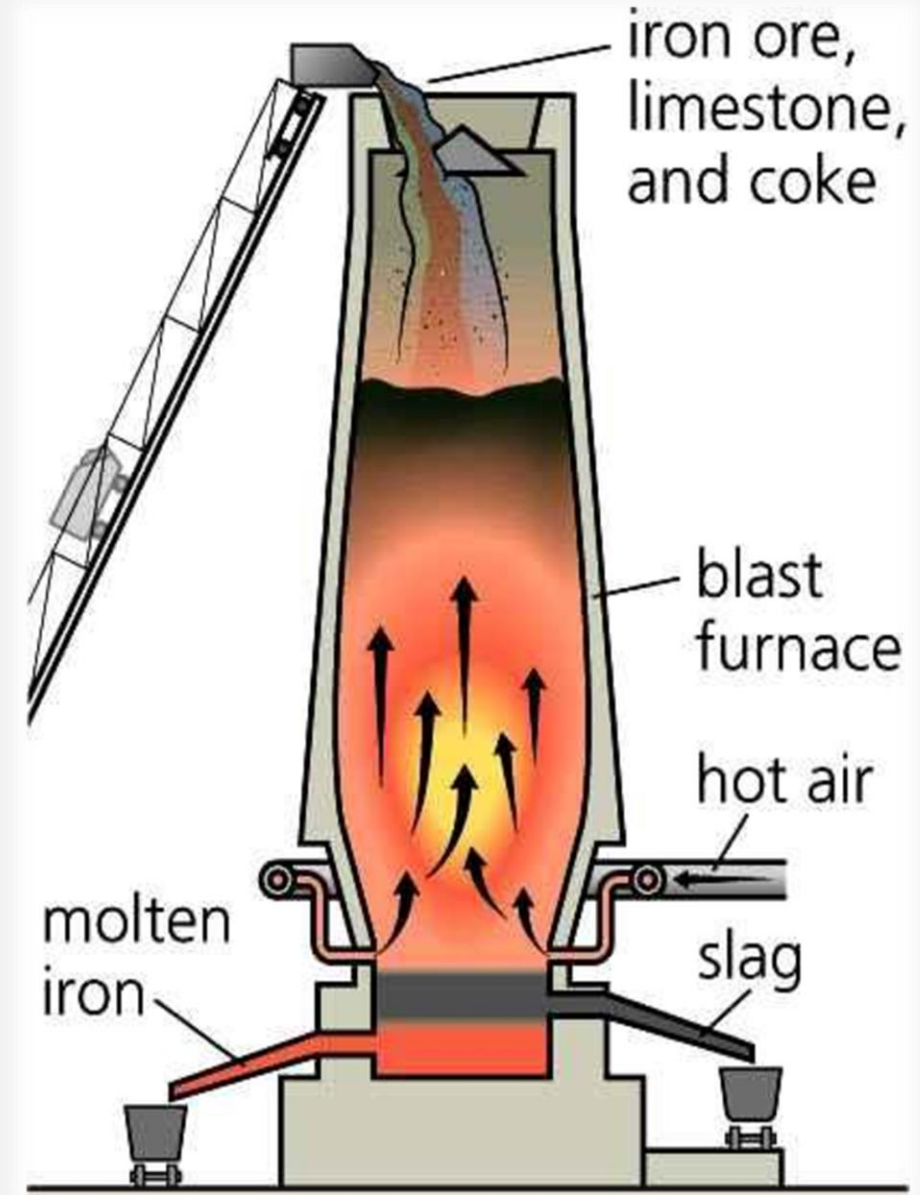


Metal Extraction

Pyrometallurgy involves high temperature processes where chemical reactions take place among gases, solids, and molten materials.

Solids containing valuable metals are treated to form intermediate compounds for further processing or converted into their elemental or metallic state.

Pyrometallurgical processes that involve gases and solids are typified by **calcining** and **roasting** operations.

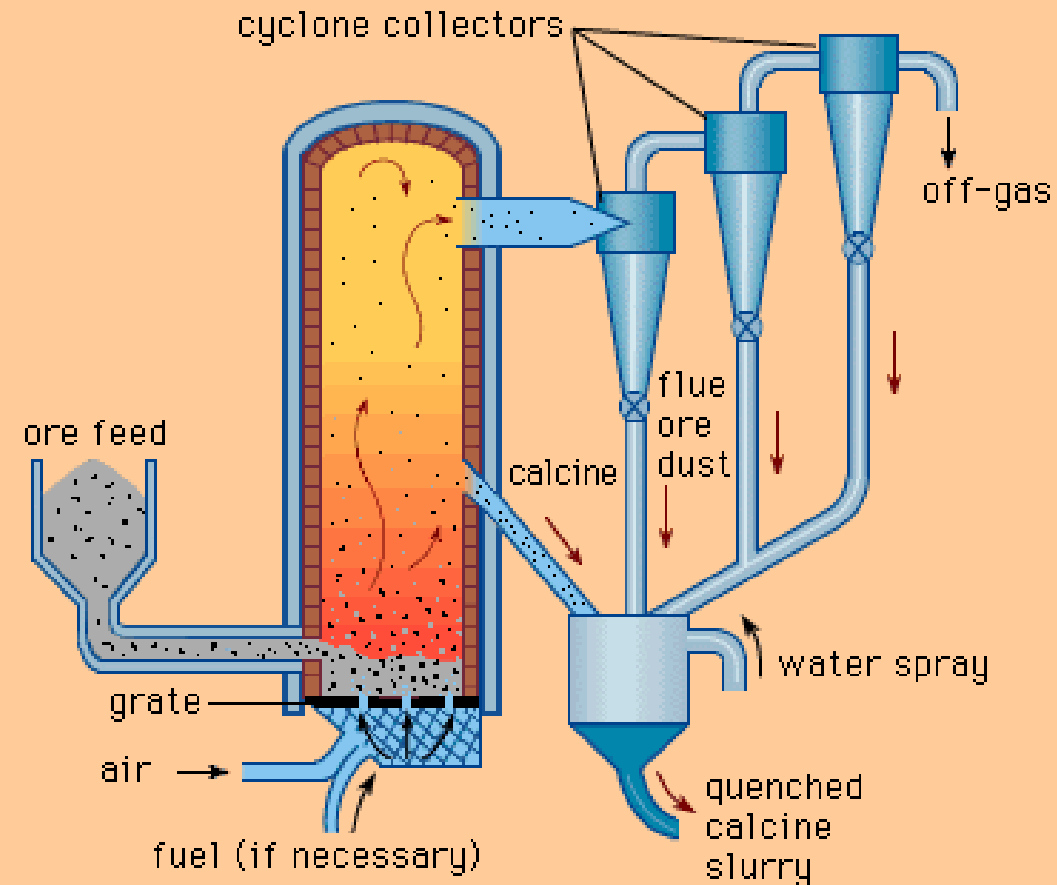
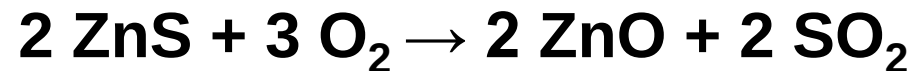
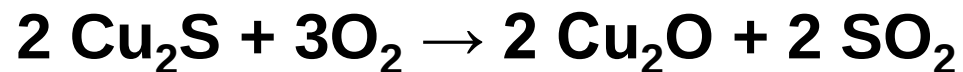


Metal Extraction

Roasting is a process of heating of concentrated ore to a **high temperature** in excess of air.

This process is generally applied to **sulphide minerals**. During roasting, the **sulphide** is converted to an oxide, and sulphur is released as sulphur dioxide, a gas.

For the ores **Cu_2S (chalcocite)** and **ZnS (sphalerite)**, balanced equations for the roasting are:

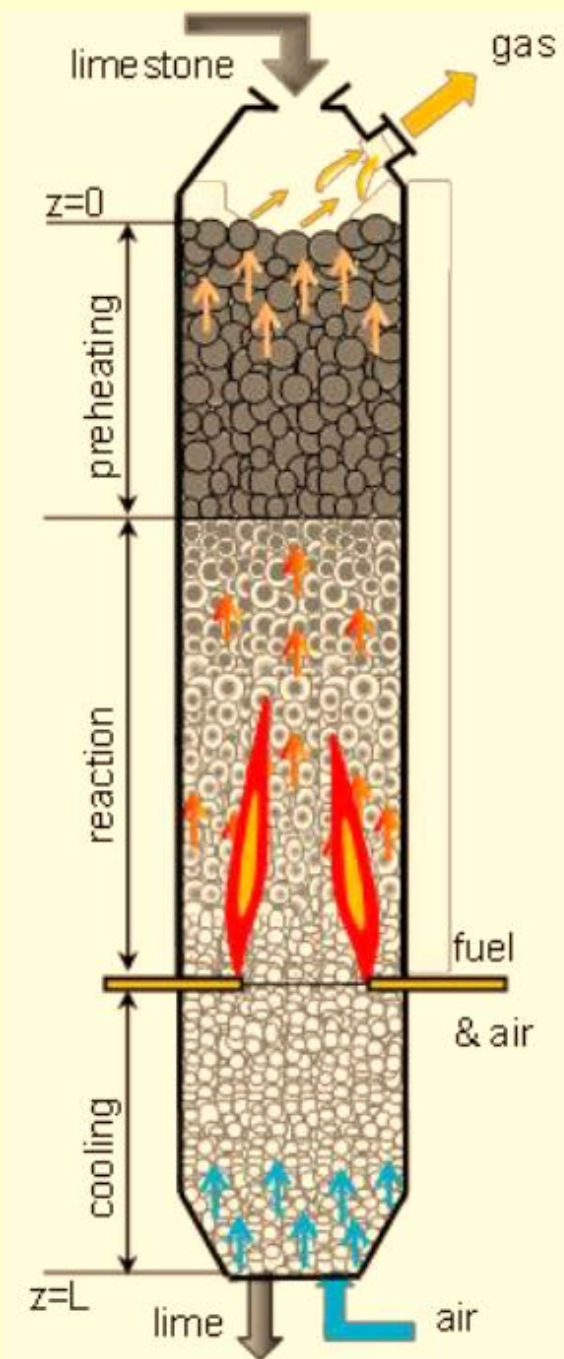


Metal Extraction

Calcination

Calcination	Roasting
1. The ore is heated in the absence of air.	1. The ore is heated in the presence of air.
2. It is used to carbonate ores.	2. It is used for sulphide ores.
3. Ex : $\text{CaCO}_3 \xrightarrow{\text{heat}} \text{CaO} + \text{CO}_2 \uparrow$ Lime stone	3. Ex : $2 \text{PbS} + 3 \text{O}_2 \xrightarrow{\text{heat}} 2 \text{PbO} + 2 \text{SO}_2$

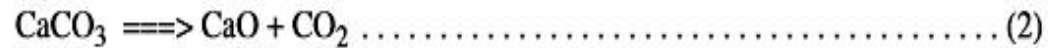
- Decomposition of some mineral compounds by high temperature
- Remove water, CO_2 and other which in the form metal hydrate and carbonate
- These ores is low temperature of decomposition



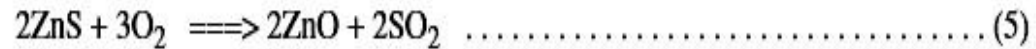
Principles of Metal Extraction

“ In pyrometallurgy, metals are extracted by converting **sulfides into oxides** and then reducing **oxides into metals**, using carbon or carbon monoxide as reducing agents. The reactions for lead, zinc, and iron are given below. ”

Lead:



Zinc:

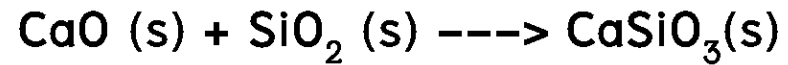


Iron:



Pyrometallurgy process

In the cassiterite ore have a impurity (SiO_2), added CaCO_3 can remove the SiO_2 in form of slag (CaSiO_3)



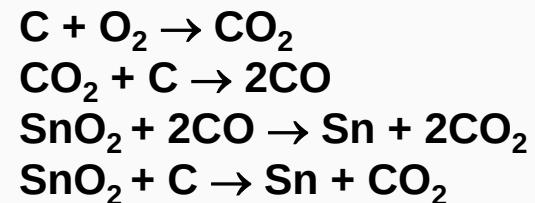
Cassiterite
(SnO_2)

Ore dressing
(SnO_2)
Concentrate

Slag

Sn melt

Reverberatory furnace
 $T \approx 1,350^\circ\text{C}$
($\text{SnO}_2 + \text{Coke} + \text{CaCO}_3$)



Tin extraction (Sn)

Cassiterite ore (SnO_2) mixed with **coke** and **calcium carbonate (CaCO_3)** in the ratio 20:4:5 into the furnace, the reaction as a followed:

Sn + Fe
impurity
(Fe < 1.0%)

Open hearth
furnace
at 232°C

FeSn_2
(solid)

Sn
(liquid)

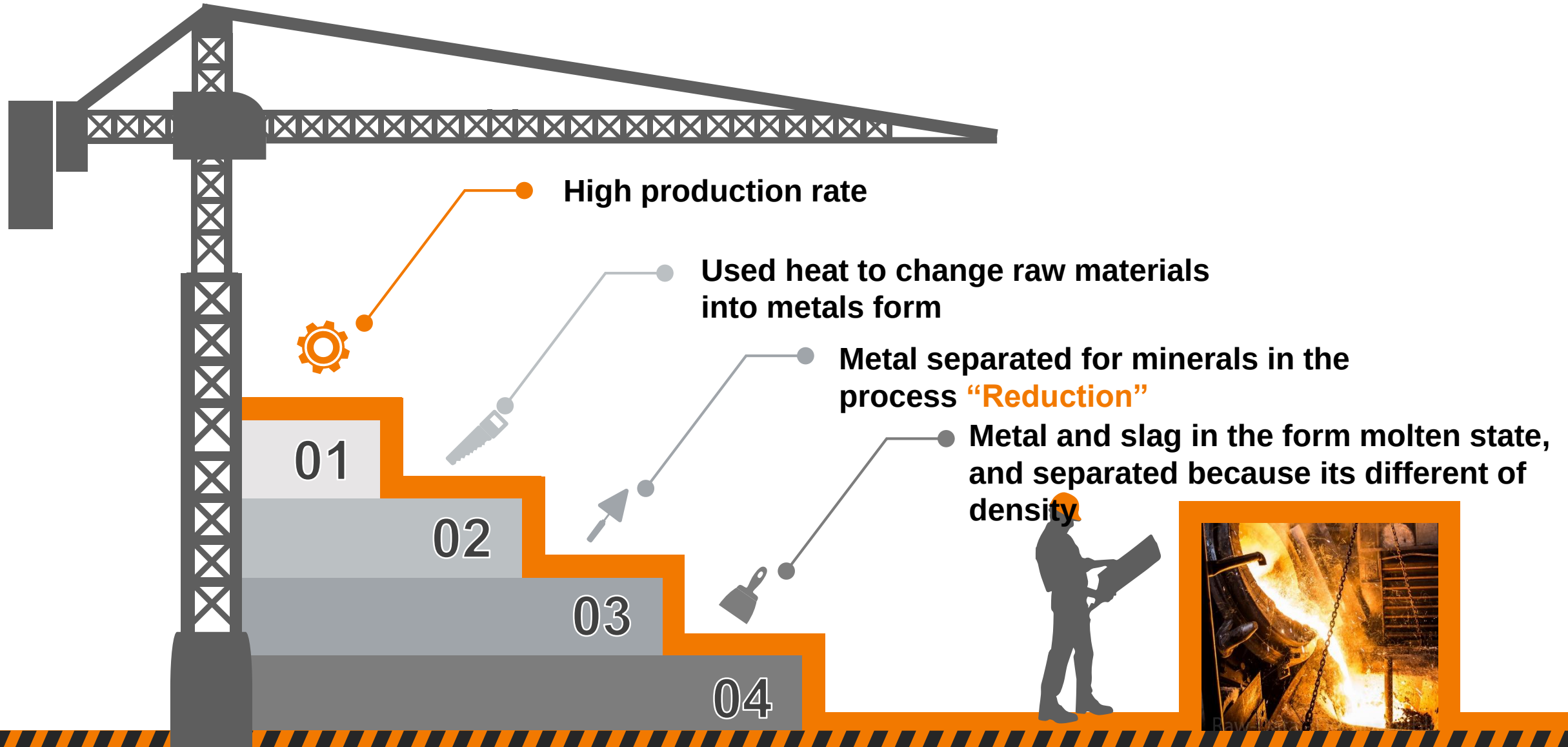
Sn + Pb
impurity

Open hearth
furnace
at 245°C + Flux
(SnCl_2)

PbCl_2
(liquid)

Sn
(liquid)

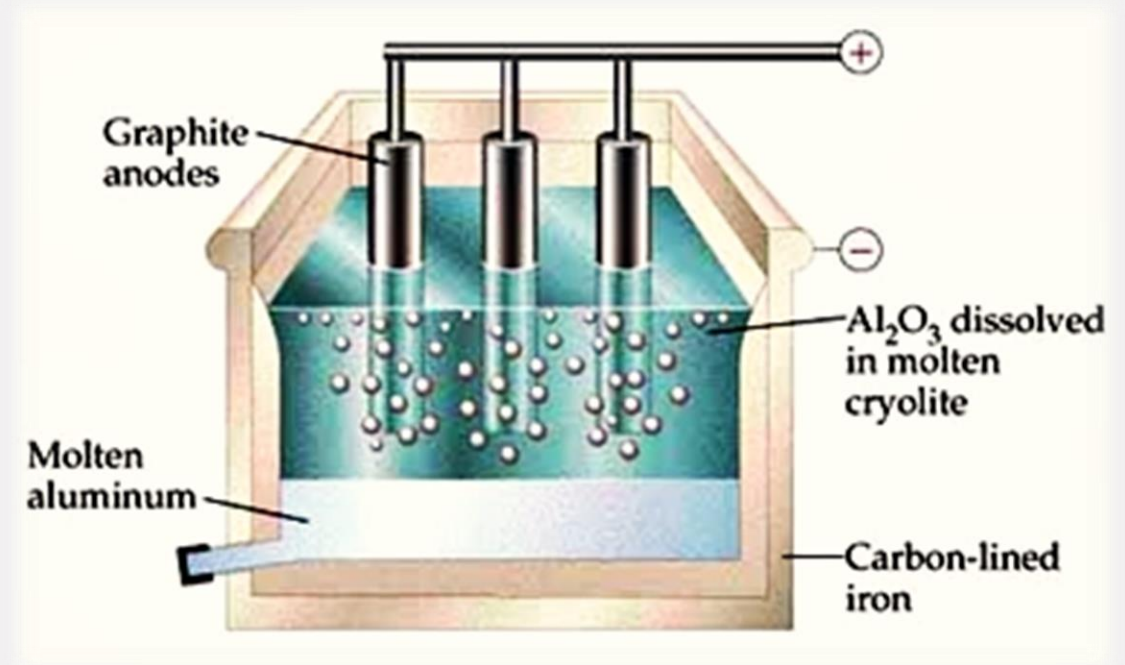
Pyrometallurgy Process



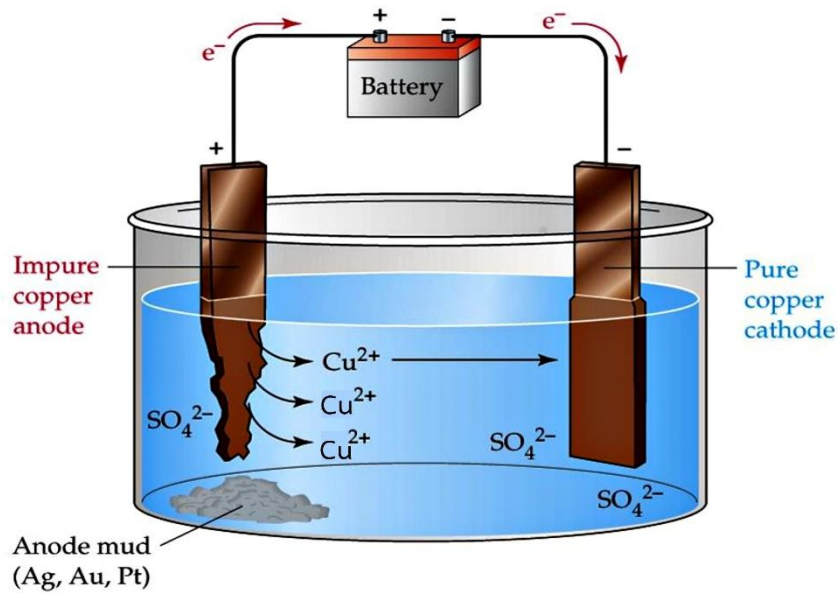
Metal Extraction

Electrometallurgy is a method that uses electrical energy to produce metals by electrolysis. Electrometallurgy is usually the last stage in metal production and is therefore preceded by pyrometallurgical or hydrometallurgical operations.

The most common types of electrometallurgical processes are **electrowinning** and **electrorefining**.



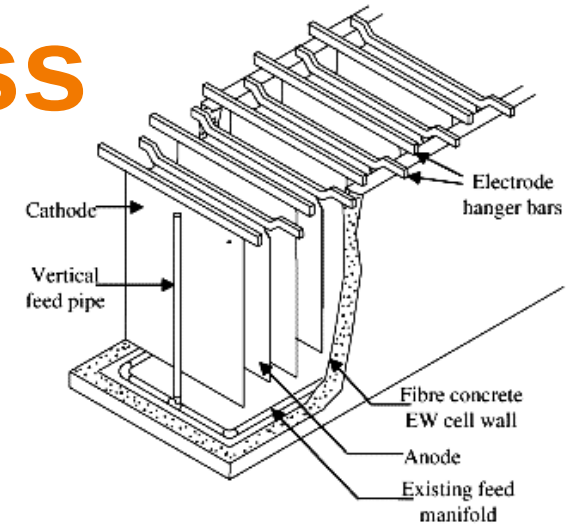
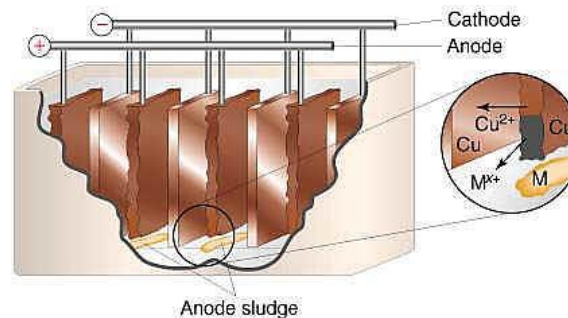
Electrometallurgical Process



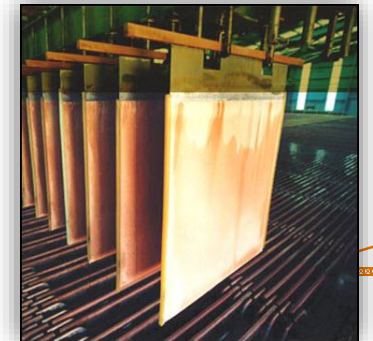
The **anodes** consist of unrefined impure metal, and as the current passes through the acidic electrolyte the anodes are corroded into the solution so that the electroplating process deposits refined pure metal onto the **cathodes**.

Electrowinning

Electrorefining

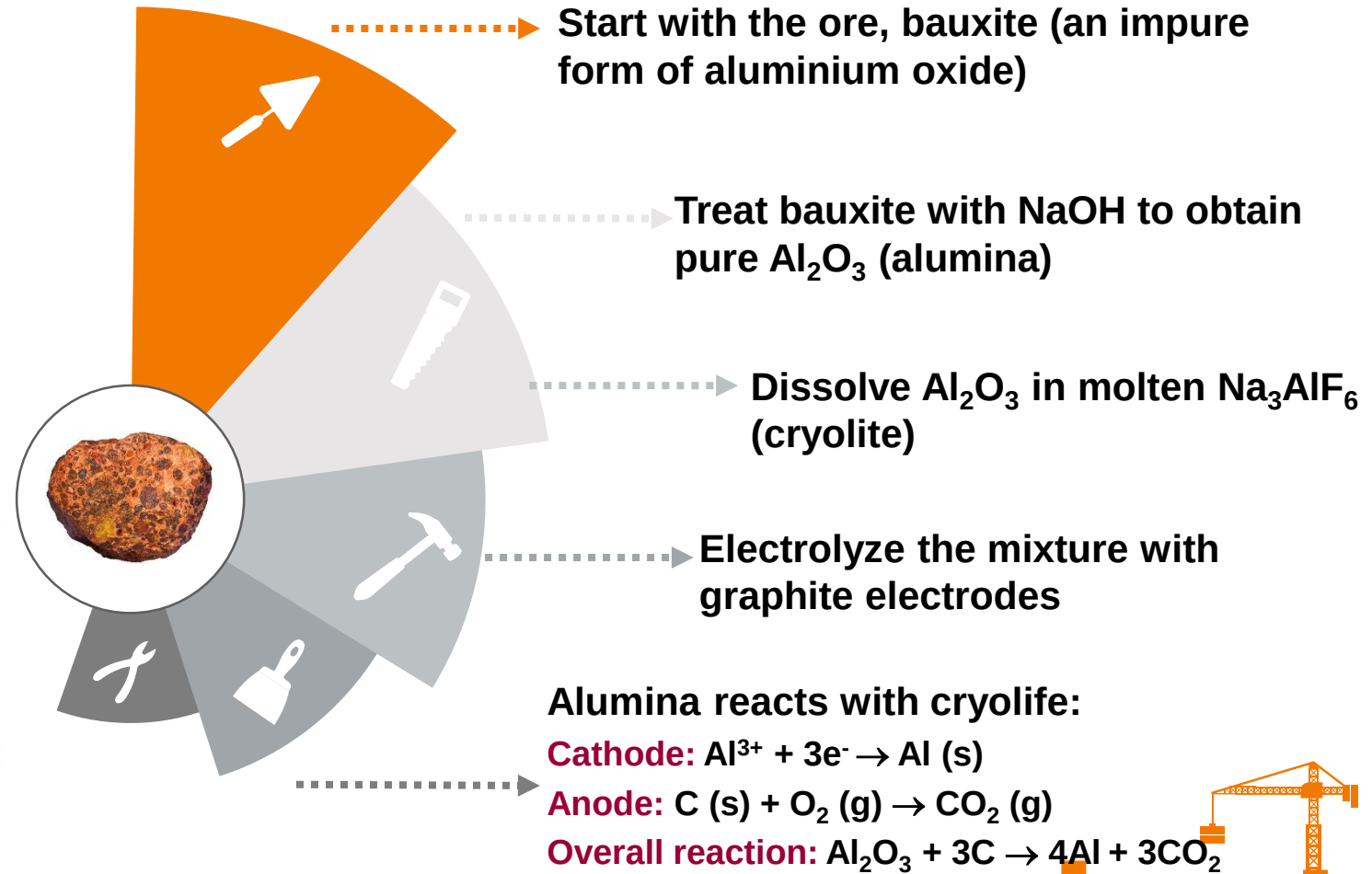
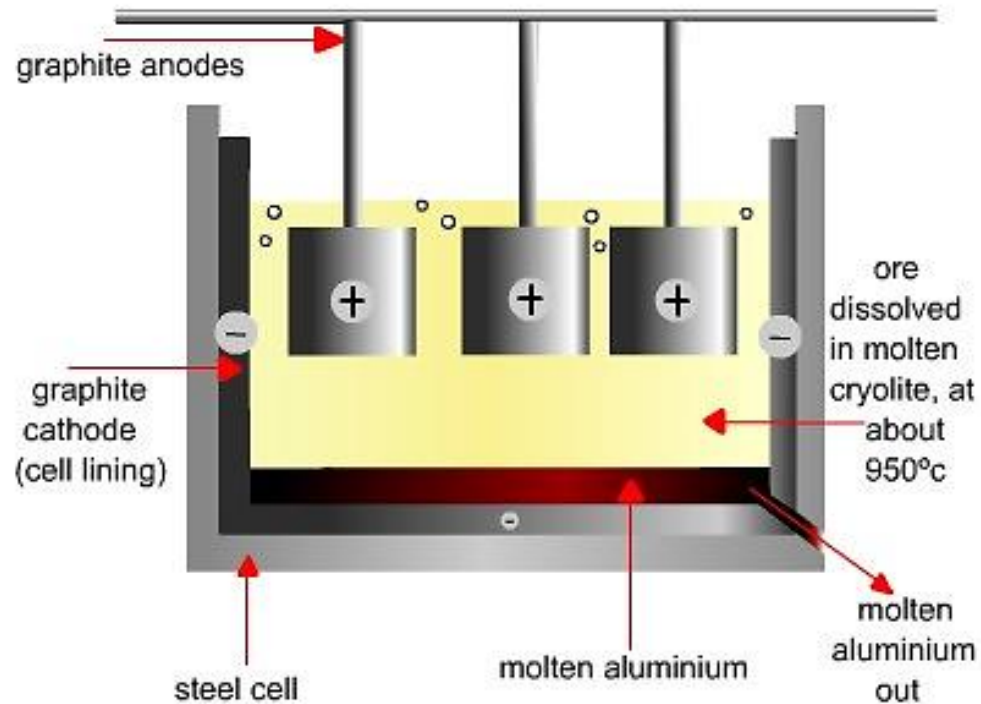


A current is passed from an inert **anode** through a liquid leach solution containing the metal so that the metal is extracted as it is deposited in an electroplating process onto the **cathode**.



Electrometallurgical Process

The Hall-Héroult Process



Metal Refining

- ❑ The process of purification of impure metals is known as **refining**.

Electrolytic Refining or Electro-refining.

This method is based on the phenomenon of electrolysis. Many metals like **copper, silver, tin, gold, zinc, nickel and chromium** are purified by electro-refining.

